

SHOPEZ – FULL STACK MERN E-COMMERCE PLATFORM

1. INTRODUCTION

Project Title:

ShopEZ – Full Stack MERN E-Commerce Platform

Team Member:

Sanjana Boddireddy

Role:

Full Stack Developer

2. PROJECT OVERVIEW

Purpose:

ShopEZ is a full-stack e-commerce web application developed using the MERN stack. The system allows users to browse products, register accounts, log in securely, add products to cart, place orders, and manage purchases through a user friendly interface.

Features:

- User Registration and Login
- Product Browsing by Categories
- Product Search Functionality
- Shopping Cart Management
- Checkout System
- Responsive User Interface
- Admin User Support
- Product Management
- MongoDB Database Integration

3. ARCHITECTURE

Frontend:

The frontend is developed using React.js and Vite. It provides an interactive and responsive user interface for customers to browse products and manage their shopping experience.

Backend:

The backend is developed using Node.js and Express.js. REST APIs are used for communication between frontend and backend.

Database:

MongoDB is used to store user information, product details, orders, and authentication data.

React Frontend



Express.js API



Node.js Server



MongoDB Database

4. SETUP INSTRUCTIONS

Prerequisites:

- Node.js
- npm
- MongoDB Community Server
- MongoDB Compass
- VS Code

Installation Steps:

1. Download the ShopEZ project.
2. Open terminal inside project folder.
3. Install backend dependencies:

```
npm install
```

4. Install frontend dependencies:

```
cd client  
npm install
```

5. Configure .env file.

6. Start backend:

```
cd server  
node index.js
```

7. Start frontend:

```
cd client
npm run dev
```

5. FOLDER STRUCTURE

Client:

- components
- pages
- assets
- App.jsx
- index.css

Server:

- config
- controllers
- middleware
- models
- routes
- index.js

6. RUNNING THE APPLICATION

Backend:

```
cd server
node index.js
```

Frontend:

```
cd client
npm run dev
```

Application URL:

<http://localhost:5173>

7. API DOCUMENTATION

Authentication APIs:

- POST /api/auth/register
- POST /api/auth/login

Product APIs:

- GET /api/products
- GET /api/products/:id

Cart APIs:

- POST /api/cart
- GET /api/cart

Order APIs:

- POST /api/orders
- GET /api/orders

Admin APIs:

- GET /api/admin/products
- GET /api/admin/orders

8. AUTHENTICATION

Authentication is implemented using JSON Web Tokens (JWT).

Features:

- User Registration
- User Login
- Token Generation
- Protected Routes
- Role-based Access Control

9. USER INTERFACE

The user interface includes:

- Home Page
- Category Section
- Product Listing Page
- Product Details Page
- Shopping Cart
- Login Page
- Registration Page
- Checkout Page
- Admin Dashboard

10. TESTING

Testing Performed:

- User Registration Testing
- User Login Testing
- Product Search Testing
- Cart Functionality Testing
- Checkout Testing

- Admin Functionality Testing
- API Endpoint Testing

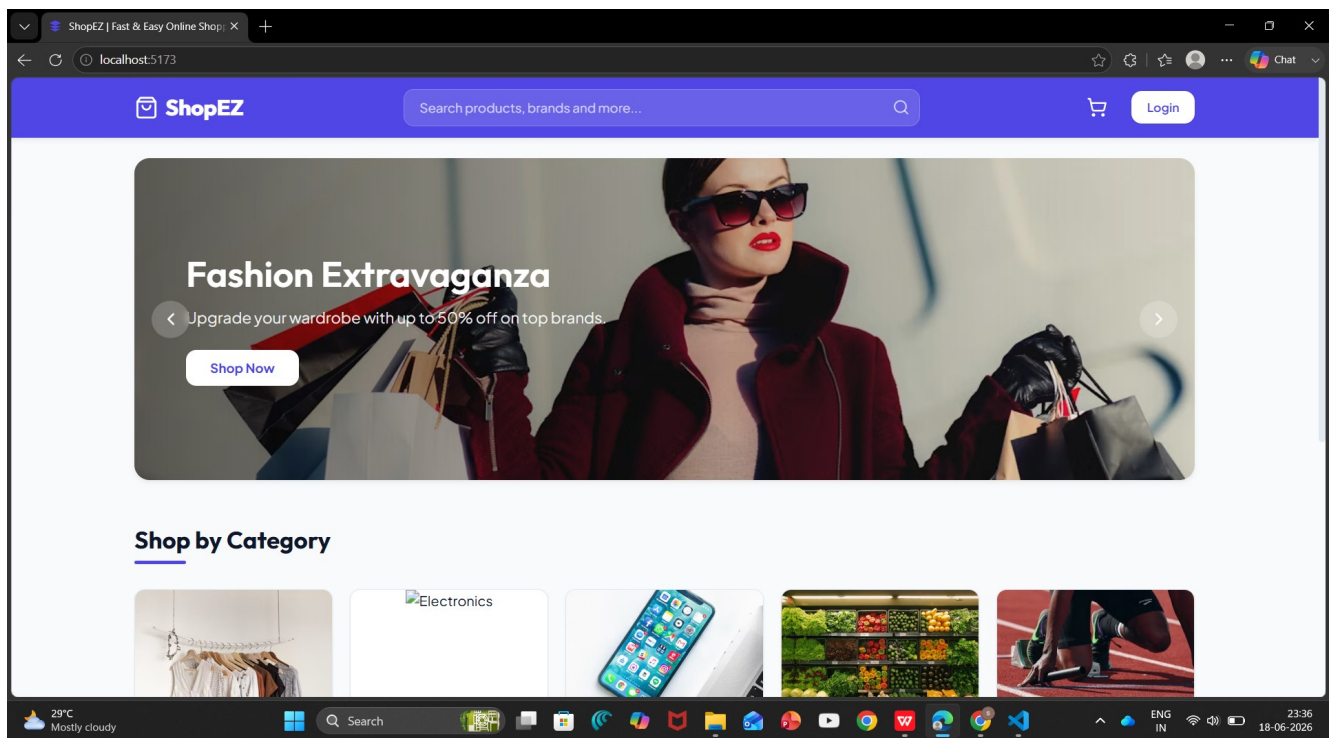
11. SCREENSHOTS OR DEMO

Demo URL:

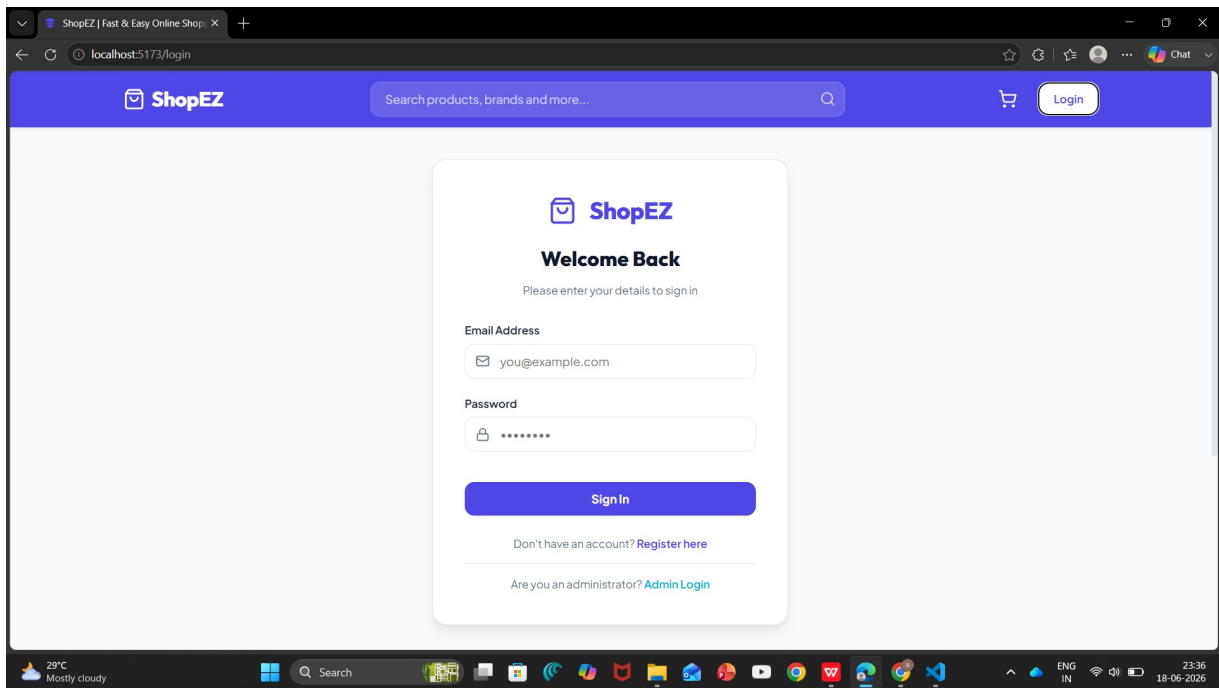
<http://localhost:5173>

Screenshots Included:

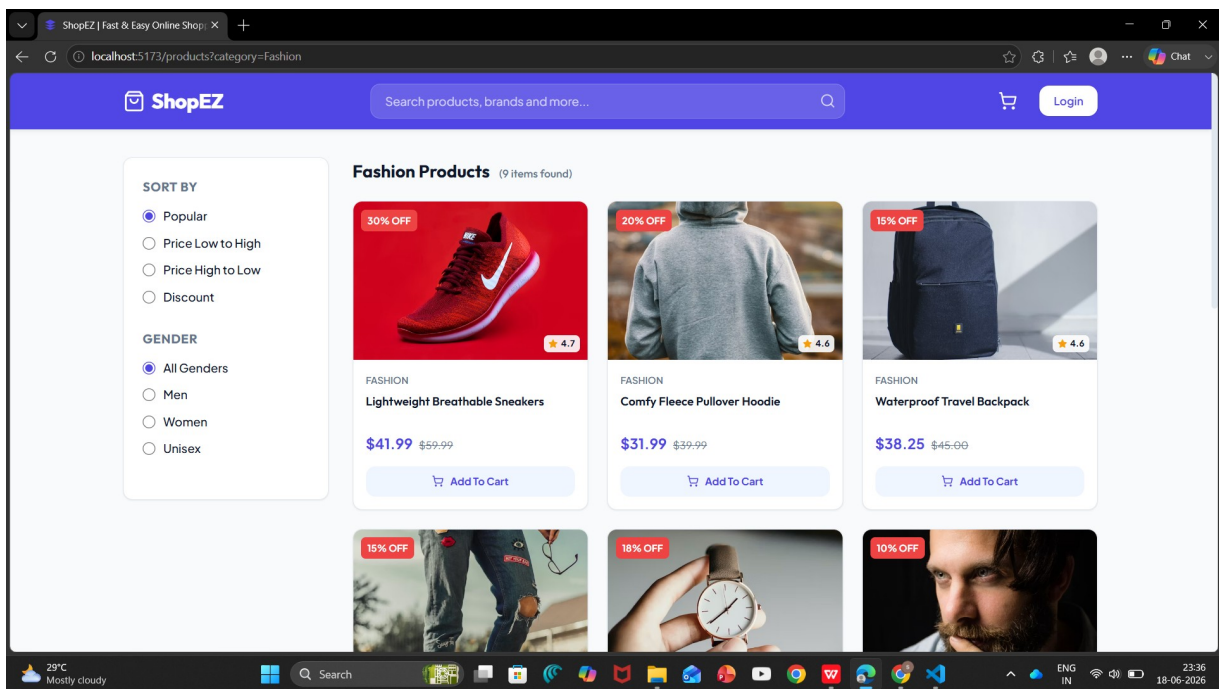
• Home Page



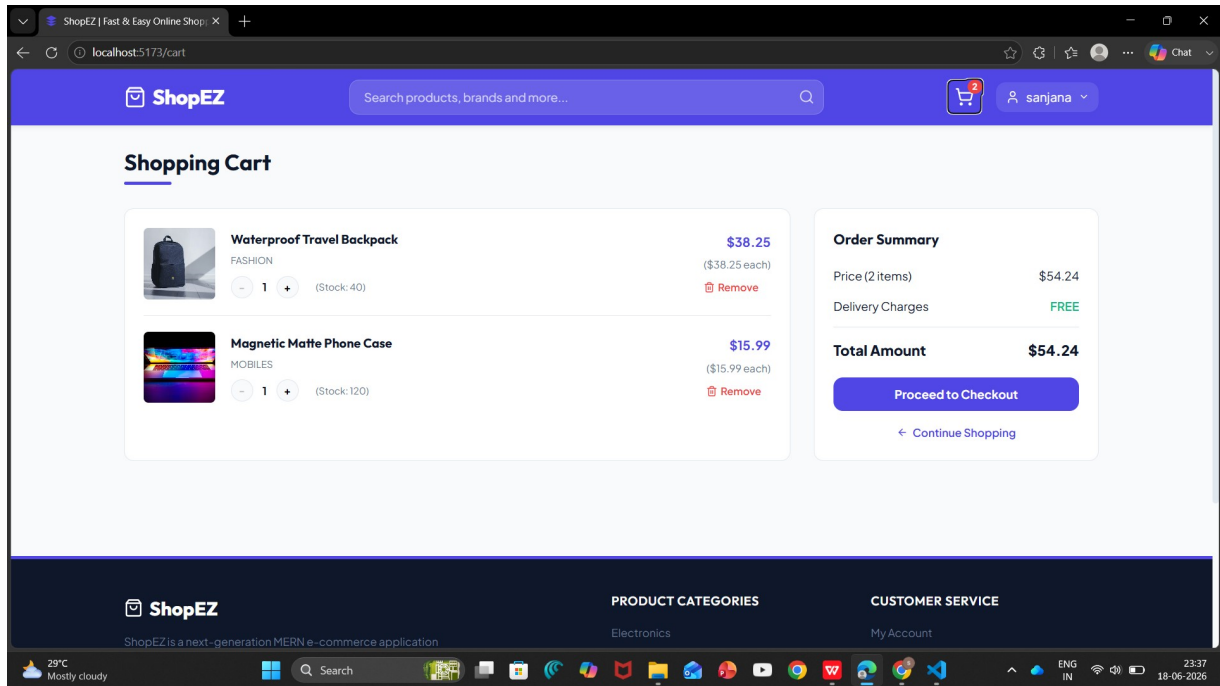
• Login Page



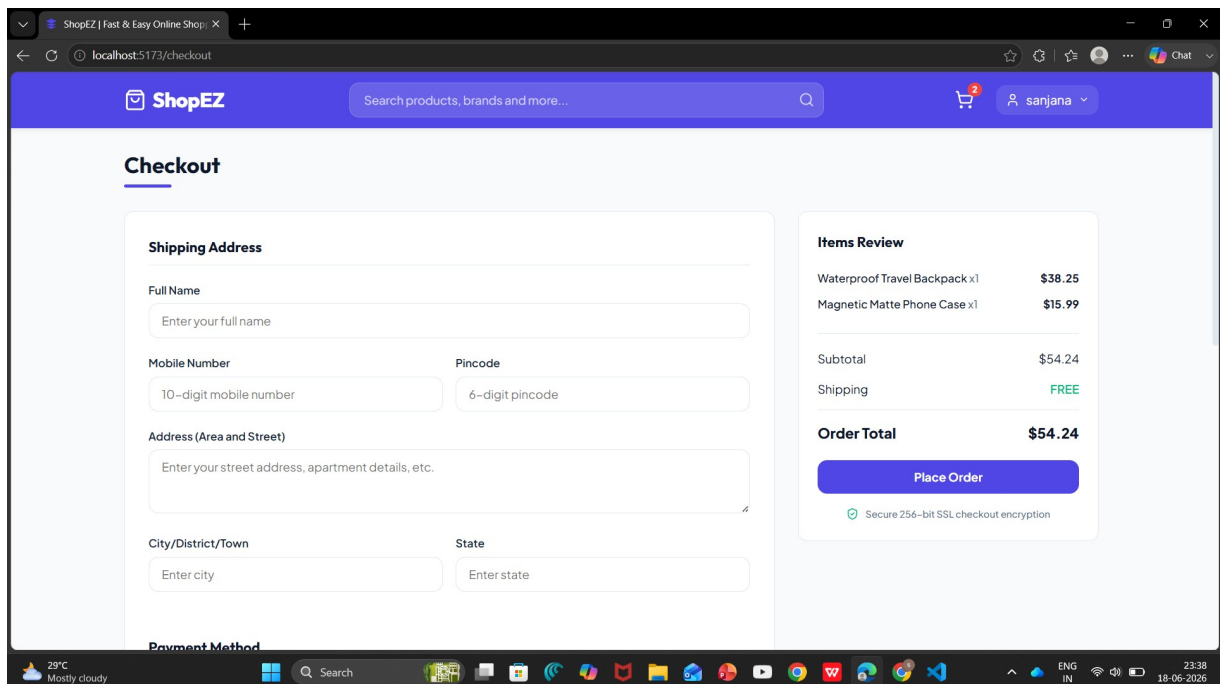
• Product Listing



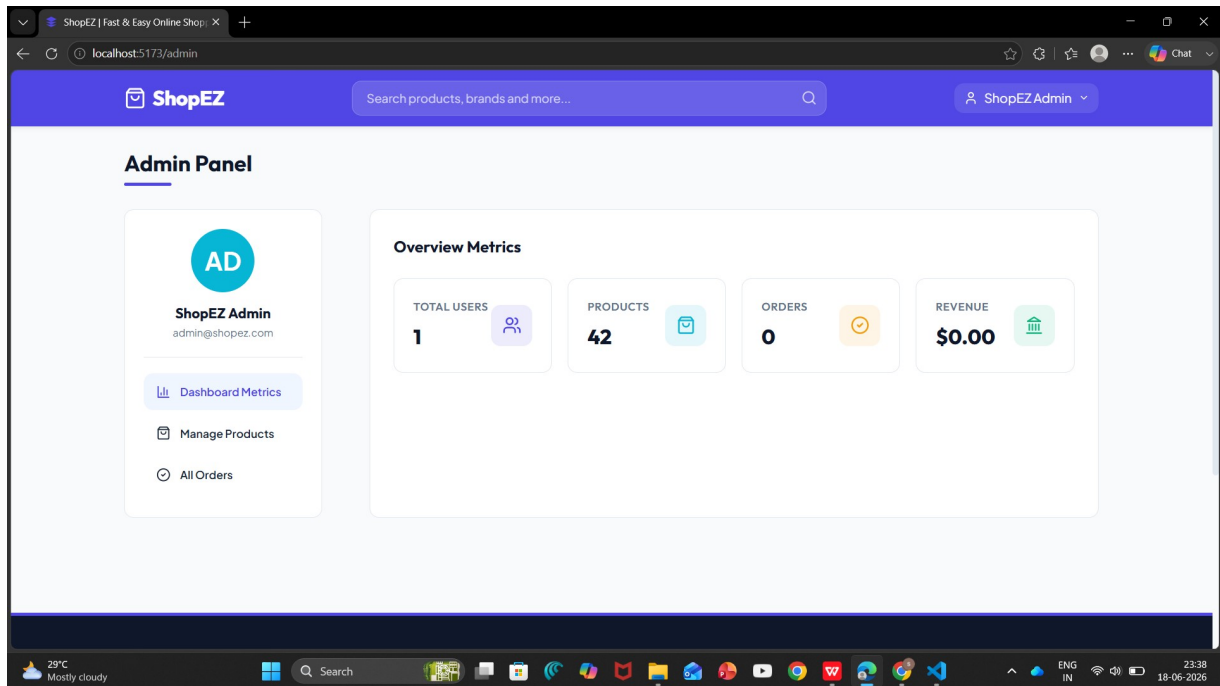
• Shopping Cart



• Checkout Page



• Admin Dashboard



12. KNOWN ISSUES

- MongoDB must be running for persistent data storage.
- Some external image URLs may occasionally fail.
- Mock Database Mode resets data after server restart.

13. FUTURE ENHANCEMENTS

- Online Payment Gateway Integration
- Wishlist Functionality
- Product Reviews and Ratings
- Order Tracking System
- Email Notifications
- Cloud Deployment
- Mobile Application Version
- AI-Based Product Recommendations